$$\text{we know that, } I = \frac{Pnr}{100}$$

$$\text{Therefore, } 6440 = \frac{46000 \times r \times \frac{7}{4}}{100}$$

$$6440 = 46000 \times r \times \frac{7}{4} \times \frac{1}{100}$$

$$r = \frac{6440 \times 4 \times 100}{46000 \times 7}$$

$$= 8\%.$$

Example 2.29 A principal becomes ₹ 10,050 at the rate of 10% in 5 years. Find the principal.

Solution

$A = ₹\,10{,}050$

$n = 5$ years

$r = 10\%$

$P = ?$

For calculating principal with the given data, we proceed as follows.
We know that,

$$I = \frac{Pnr}{100}$$

$$A = P + I$$

$$A = P + \frac{Pnr}{100}$$

$$A = P\left(1 + \frac{nr}{100}\right)$$

$$\text{Therefore, } 10,050 = P\left(1 + \frac{10 \times 5}{100}\right)$$

$$= P\left(1 + \frac{50}{100}\right)$$

$$= P\left(\frac{150}{100}\right)$$

$$= P\left(\frac{3}{2}\right)$$

$$\text{Therefore, } P = 10{,}050 \times \frac{2}{3} = 3350$$

Hence, Principal = ₹ 3,350.

Think

In simple interest, a sum of money doubles itself in 10 years. In how many years it will get triple itself.

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